[10, 20, 30, 50, 10, 70, 30]

SZ = 3.

0	1	2	3	4	5	6
1	2	3	4	5	6	7
50	30	20				10
70	30					10
70	30	20	10	10	10	10
70.	30	20	$-\infty$	$-\infty$	$-\infty$	10

Prev. Sm → O(n).

next Sm → O(n)

TC = O(n).

AS = O(n).

	0	1	2	3	4	5	6.
-1	10	20	30	50	10	70	30.
PS = -1		0	1	2	-1	4	4
NS = 7		4	4	4	7	6	7
WS = 7	3	2	1	7	1	2	

window size = NS[i] - PS[i] - 1.

0	1	2	3	4	5	6
$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
70	30	20	10.	10	10	10

```
vector<int> maxSlidingWindow(vector<int>&nums, int k) {

    vector<int> prevSmaller = getPrevSmaller(arr);
    vector<int> nextSmaller = getNextSmaller(arr);

    int n = arr.size();
    vector<int> result(n, INT_MIN);

    for (int i = 0; i < n; i++) {
        int window_size = nextSmaller[i] - prevSmaller[i] - 1;
        result[window_size - 1] = max(result[window_size - 1], arr[i]);
    }

    for (int i = n - 2; i >= 0; i--) {
        result[i] = max(result[i], result[i + 1]);
    }


    return result;
}
```